Sleep Duration, Sleep Regularity and Final Exam Performance in Undergraduates

Synthetic sample study. The university, participants and results are fictional.

Abstract

We followed 212 undergraduates at Northfield University for the six weeks before their final exam in an introductory chemistry course, measuring sleep with wrist-worn actigraphs. Each additional hour of mean nightly sleep was associated with a **4.1 percentage-point** higher final exam score after controlling for prior grade point average. Students whose bedtimes varied more scored lower, independently of how long they slept. Sleeping the night before the exam was not associated with scores once the six-week average was taken into account.

Introduction

Students commonly trade sleep for study time near exams. Laboratory studies show that sleep supports memory consolidation, but evidence from real courses, measured objectively over weeks rather than by self-report, is limited. We asked whether average sleep duration and sleep regularity across a whole exam preparation period predict exam performance.

Methods

Participants

We enrolled 231 students from CHEM-101 in the autumn semester. Nineteen were excluded for wearing the actigraph less than 80% of the time, leaving **212 participants** (118 women and 94 men; mean age 20.3 years). Participants received course credit for taking part. The protocol was approved by the institutional review board under protocol number IRB-2023-117.

Sleep measurement

Participants wore an ActiTrack 3 wrist actigraph continuously for 42 days before the final exam and kept a brief daily sleep diary, which was used to mark when the device was removed. Sleep duration was the total sleep time per night; **sleep regularity** was the standard deviation of sleep onset time across nights.

Outcome and analysis

The outcome was the final exam score as a percentage, graded blind to sleep data. We used linear regression with mean sleep duration and sleep-onset variability as predictors, controlling for prior GPA, gender and age. Analyses were run in R 4.3.

Results

Sleep in the sample

Mean nightly sleep was **6.8 hours** (SD 0.9). Thirty-one percent of participants averaged less than six hours a night, and sleep fell by a mean of 38 minutes in the final week before the exam.

Sleep duration and scores

Each additional hour of mean nightly sleep predicted a final exam score **4.1 points higher** (95% confidence interval 2.6 to 5.6; $p < 0.001$), controlling for prior GPA, gender and age.

Mean exam score by quartile of sleep duration:

- Shortest quartile (under 6.1 hours): 68.2%
- Second quartile: 72.5%
- Third quartile: 75.9%
- Longest quartile (over 7.5 hours): 78.4%

Sleep regularity

Independently of duration, each one-hour increase in the standard deviation of sleep onset predicted a score **2.3 points lower** (95% CI 0.9 to 3.7).

The night before the exam

Sleep on the final night alone was not associated with scores once the six-week average was included ($p = 0.41$).

Discussion

Across six weeks of objectively measured sleep, students who slept longer and on a more regular schedule scored higher on their final exam, and the association held after accounting for prior academic performance. The difference between the shortest and longest sleepers, about ten percentage points, is roughly one letter grade.

Regularity matters

The independent effect of sleep regularity suggests that a consistent schedule matters in addition to total sleep. Irregular bedtimes shift the body clock, which can reduce sleep quality even when the hours add up.

Cramming

That a single night before the exam did not predict scores suggests that last-minute "all-nighters" do not help, and that what matters is sleep across the whole period in which material is learned, consistent with the role of sleep, particularly slow-wave sleep, in consolidating memories formed earlier.

Limitations and Conclusion

Limitations

- The study is observational: sleep was not assigned, so we cannot conclude that more sleep causes higher scores. Conscientious students may both sleep more regularly and study more effectively.
- All participants took one course at one university, so the results may not generalise.
- Actigraphy tends to overestimate sleep compared with laboratory polysomnography.
- Participation was voluntary, so the sample may differ from non-participants.

Conclusion

Students preparing for exams should aim for at least seven hours of sleep a night on a consistent schedule throughout the term, rather than catching up before the exam. A randomised trial of a sleep extension intervention is the natural next step.

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